Why Coding? A one-page guide for families

Kids Coding · 3 levels · 10–12 weeks each · one 60-minute session a week · starts January 2027

What the research says

- **Strong** Children learn to program; problem solving and planning improve (Montuori et al., 2024; Scherer et al., 2019).
- **Mixed** Math and creativity gains shrink in stricter studies; reading and grades benefit least (Scherer et al., 2019).
- **Mixed** Blocks before text: no lasting harm or edge (Xu et al., 2019).
- **Open** General persistence; long-term effects of AI (Kazemitabaar et al., 2023).

Our promise: capable, confident beginner programmers. No promises about grades, jobs or college.

How we teach, and why

Real projects in games and art (Resnick et al., 2009); pair programming (Umapathy & Ritzhaupt, 2017); a bug hunt every week (Michaeli & Romeike, 2019); unplugged and predict-then-run activities (Sentance et al., 2019); ideas that return, deeper, each level.

How you can help (no coding needed)

- **Ask for a demo:** “Show me what you made.”
- **Ask about the bug,** not the score: “What went wrong? How did you find it?”
- **Praise strategies and effort,** not “being a natural.”
- **Protect 15 minutes** a week for homework, at a regular time.
- **Encourage everyone equally:** girls hear less encouragement than boys.
- **Come to the showcase** at the end of each level.

The three levels

1. **Introduction** (Scratch blocks, then Python turtle): steps in order, loops, events and choices, variables, first Python. *Project:* My Interactive Art Show.
2. **Intermediate** (Python): functions, lists and dictionaries, files, testing, keyboard-driven games. *Project:* Arcade Game or Art Studio.
3. **Advanced** (Python projects + git): objects and classes, search and sort, recursion, data, how simple AI learns. *Project:* Build Something Big.

Each level takes 10–12 weeks (30–36 weeks for all three). Checkpoint in week 6, showcase in week 12; rubrics guide support, no grades.

A typical session

Group: up to 4 children, working in 2 pairs. **Homework:** 15–20 minutes a week, optional but encouraged. **Equipment:** a Mac, the free Scratch app and Thonny; no accounts, no internet during lessons.

Digital citizenship & safety: our promises

- **No accounts, no logins, no email addresses.**
- **Nicknames only;** programs never ask for personal information.
- **Nothing is uploaded;** projects stay on the computers in the room.
- **Screen breaks every session,** plus unplugged activities.
- **Color-blind-safe materials** and a dyslexia-friendly edition.

Questions? [Presenter name] · [email or phone]

Full references and study notes: appendix of the slide deck.